Crown Change Equations for CONUS

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Data

We took the FIA crown change data subset for all species, filtered out missing data and trees with negative height to live crown change ($\Delta htlc$). This left 2389711 observations.

The observations are distributed among states as shown in Table 1.

Table 1: Height to Live Crown Change Observations by State

State	n
MN	178509
WI	167983
OR	161744
MI	142694
ME	141069
GA	123949
WA	98456
AL	92972
CA	88648
MS	79191
VA	76917
NY	75459
SC	72181
NC	72008
PA	69426
AR	67303
MO	65656
TN	62495
TX	52452
MT	49059
KY	48974

Table 1: Height to Live Crown Change Observations by State

State	n
WV	38612
NH	37157
ID	33557
FL	33066
CO	29436
LA	28369
VT	27541
OH	25433
OK	22068
IN	19950
IL	14803
MA	12053
UT	11570
IA	8550
AZ	7907
NJ	7256
MD	6585
KS	6184
SD	6085
NM	5645
CT	5245
WY	4465
NE	3018
RI	2450
ND	2403
DE	2147
NV	1011

The observations are distributed among species as shown in Table 2.

Table 2: Height to Live Crown Change Observations by Species

SPCD	CommonName	n
131	loblolly pine	255274
316	red maple	157915
202	Douglas-fir	130560
318	sugar maple	88218
746	quaking aspen	85387

Table 2: Height to Live Crown Change Observations by Species

SPCD	CommonName	n
12	balsam fir	74582
611	sweetgum	59878
802	white oak	59211
122	ponderosa pine	56698
108	lodgepole pine	56382
241	northern white-cedar	51515
129	eastern white pine	42637
621	yellow-poplar	42299
833	northern red oak	39794
125	red pine	37697
111	slash pine	36470
261	eastern hemlock	36128
531	American beech	33918
762	black cherry	30825
97	red spruce	29543
95	black spruce	29496
263	western hemlock	28604
110	shortleaf pine	28590
832	chestnut oak	28523
375	paper birch	26969
19	subalpine fir	26032
835	post oak	24855
837	black oak	23940
68	eastern redcedar	23829
15	white fir	23473
827	water oak	23283
371	yellow birch	22951
693	blackgum	21993
543	black ash	21837
541	white ash	20550
17	grand fir	19422
93	Engelmann spruce	18364
951	American basswood	18207
743	bigtooth aspen	17170
972	American elm	16948
71	tamarack (native)	16783
544	green ash	16575
403	pignut hickory	15466
409	mockernut hickory	15069

Table 2: Height to Live Crown Change Observations by Species

SPCD	CommonName	n
11	Pacific sliver fir	14970
711	sourwood	14572
121	longleaf pine	14560
372	sweet birch	14111
132	Virginia pine	13614
94	white spruce	12916
971	winged elm	12740
812	southern red oak	12431
806	scarlet oak	12364
105	jack pine	12128
242	western redcedar	10967
407	shagbark hickory	10488
631	tanoak	10146
701	eastern hophornbeam	9890
823	bur oak	9692
264	mountain hemlock	8059
81	incense-cedar	7666
602	black walnut	7127
73	western larch	6745
694	swamp tupelo	6463
931	sassafras	6422
402	bitternut hickory	6368
820	laurel oak	6310
741	balsam poplar	5921
408	black hickory	5835
591	American holly	5828
211	redwood	5654
462	hackberry	5598
491	flowering dogwood	5497
351	red alder	5474
391	American hornbeam, musclewood	5168
653	sweetbay magnolia	5129
809	northern pin oak	5089
20	California red fir	5033

Proposed Model

The model we propose to estimate $\Delta htlc$ is Equation 1.

$$\Delta htlc = cl(1 - e^{(\beta_0 + \beta_1 \Delta ht + \beta_2 cch)}) \quad (1)$$

where:

- $\Delta htlc$: change in height to live crown (feet)
- Δht : height growth (feet)
- cl : crown length (feet)
- cch : crown closure at tree tip (fraction of acre)

The rationale for the model form is:

- Change in height to live crown is bound by `cl` which is the maximum change in `htlc` that is possible (the tree loses its entire crown).
- Change is driven by current height growth (Δht) which was found to be strongly correlated and likely to be associated with crown dynamics (both building and receding crown) and by `cch` which is a measure of the competition the tree sees in its peers and superiors.

Our expectation is as Δht and `cch` increase, $\Delta htlc$ will increase.

We used an integrative fitting approach to account for varying period lengths.

Table 3: Parameter Estimates

SPCD	CommonName	n	b0	b1	b2	rmse
131	loblolly pine	255274	-0.0098379	-0.0407301	-0.0816070	4.905528
316	red maple	157915	-0.0214006	-0.0196690	-0.0230880	4.326861
202	Douglas-fir	130560	-0.0103906	-0.0154542	-0.0466045	8.534247
318	sugar maple	88218	-0.0203143	-0.0162711	-0.0172887	4.550948
746	quaking aspen	85387	-0.0288695	-0.0272429	-0.0308612	4.004679
12	balsam fir	74582	-0.0407217	-0.0103336	-0.0061238	4.199550
611	sweetgum	59878	-0.0300934	-0.0209799	-0.0290264	5.031961
802	white oak	59211	-0.0236340	-0.0156086	-0.0273967	4.940420
122	ponderosa pine	56698	-0.0124210	-0.0098792	-0.0342991	5.784173

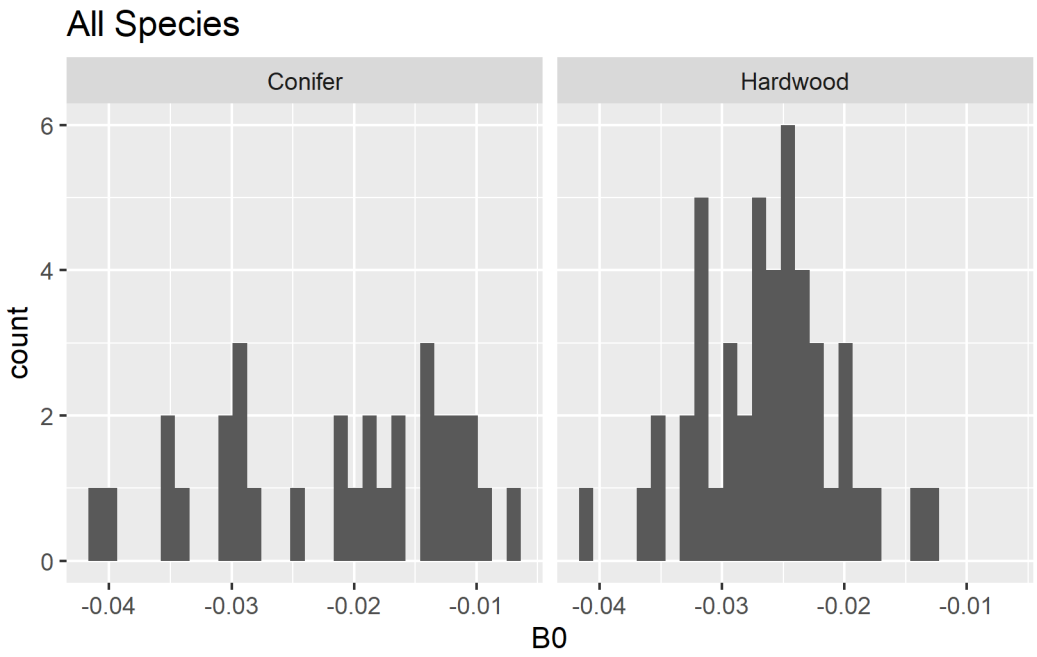
SPCD	CommonName	n	b0	b1	b2	rmse
108	lodgepole pine	56382	-0.0162559	-0.0163466	-0.0183370	4.670654
241	northern white-cedar	51515	-0.0303075	-0.0236978	-0.0068666	3.722421
129	eastern white pine	42637	-0.0277472	-0.0161392	-0.0329088	4.882960
621	yellow-poplar	42299	-0.0265923	-0.0172789	-0.0338202	5.707368
833	northern red oak	39794	-0.0196792	-0.0178678	-0.0291396	4.597741
125	red pine	37697	-0.0290298	-0.0293058	-0.0240417	3.902059
111	slash pine	36470	-0.0210122	-0.0391646	-0.0819294	4.478701
261	eastern hemlock	36128	-0.0243909	-0.0124682	-0.0079695	4.784762
531	American beech	33918	-0.0319743	-0.0123362	-0.0067374	5.222581
762	black cherry	30825	-0.0264433	-0.0273695	-0.0357791	4.439462
97	red spruce	29543	-0.0288757	-0.0188649	-0.0125133	3.428453
95	black spruce	29496	-0.0399102	-0.0191860	-0.0042322	4.156381
263	western hemlock	28604	-0.0115747	-0.0123705	-0.0146602	8.050193
110	shortleaf pine	28590	-0.0195922	-0.0324231	-0.0485041	3.971511
832	chestnut oak	28523	-0.0239460	-0.0175716	-0.0261901	4.552422
375	paper birch	26969	-0.0275348	-0.0241519	-0.0239150	3.993965
19	subalpine fir	26032	-0.0191910	-0.0006758	-0.0106562	5.621828
835	post oak	24855	-0.0243296	-0.0207106	-0.0261630	3.718509
837	black oak	23940	-0.0244112	-0.0185073	-0.0300784	4.639492
68	eastern redcedar	23829	-0.0292518	-0.0232136	-0.0168281	3.955748

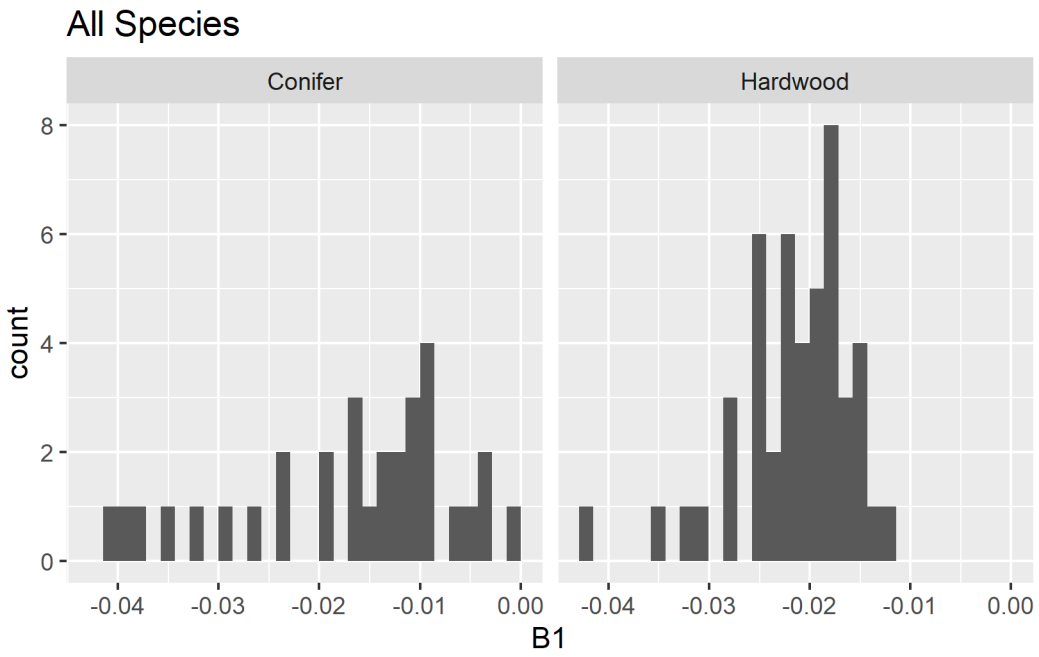
SPCD	CommonName	n	b0	b1	b2	rmse
15	white fir	23473	-0.0158597	-0.0052144	-0.0224256	7.318469
827	water oak	23283	-0.0280724	-0.0191311	-0.0329537	5.139742
371	yellow birch	22951	-0.0199775	-0.0185009	-0.0172167	4.020872
693	blackgum	21993	-0.0298993	-0.0225361	-0.0194820	4.163388
543	black ash	21837	-0.0356011	-0.0247788	-0.0220601	4.113118
541	white ash	20550	-0.0297827	-0.0208983	-0.0290710	4.869206
17	grand fir	19422	-0.0184967	-0.0040720	-0.0227532	7.441703
93	Engelmann spruce	18364	-0.0138317	-0.0039041	-0.0185537	5.769686
951	American basswood	18207	-0.0249093	-0.0196465	-0.0230942	4.475166
743	bigtooth aspen	17170	-0.0251245	-0.0249537	-0.0458886	4.516001
972	American elm	16948	-0.0331060	-0.0232407	-0.0139086	4.234021
71	tamarack (native)	16783	-0.0341860	-0.0167426	-0.0380040	4.683868
544	green ash	16575	-0.0315506	-0.0224050	-0.0279948	4.755378
403	pignut hickory	15466	-0.0259062	-0.0152976	-0.0237725	5.072553
409	mockernut hickory	15069	-0.0256740	-0.0153625	-0.0252326	4.677533
11	Pacific sliver fir	14970	-0.0140409	-0.0089437	-0.0171612	6.106812
711	sourwood	14572	-0.0270620	-0.0321758	-0.0160424	3.629018
121	longleaf pine	14560	-0.0208781	-0.0375949	-0.0620906	4.487638
372	sweet birch	14111	-0.0171376	-0.0244426	-0.0175905	4.401568
132	Virginia pine	13614	-0.0306587	-0.0350020	-0.0441767	4.505205

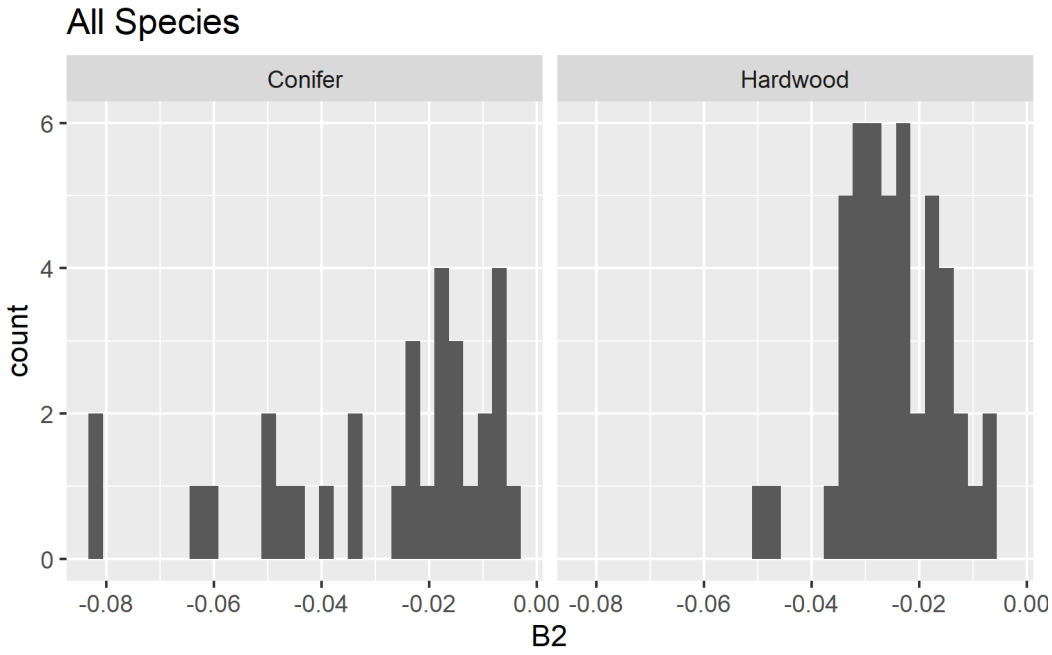
SPCD	CommonName	n	b0	b1	b2	rmse
94	white spruce	12916	-0.0348806	-0.0142850	-0.0153721	4.952566
971	winged elm	12740	-0.0329533	-0.0221849	-0.0165849	3.943727
812	southern red oak	12431	-0.0249622	-0.0189760	-0.0278618	4.939529
806	scarlet oak	12364	-0.0228741	-0.0174352	-0.0305461	4.808942
105	jack pine	12128	-0.0355247	-0.0266457	-0.0510927	3.754123
242	western redcedar	10967	-0.0106679	-0.0063642	-0.0149897	6.902488
407	shagbark hickory	10488	-0.0234159	-0.0169122	-0.0209049	4.723846
631	tanoak	10146	-0.0134099	-0.0224215	-0.0159310	4.844989
701	eastern hophornbeam	9890	-0.0257512	-0.0253129	-0.0127936	3.676321
823	bur oak	9692	-0.0311641	-0.0150296	-0.0268124	4.335401
264	mountain hemlock	8059	-0.0122030	-0.0091913	-0.0058222	5.836419
81	incense-cedar	7666	-0.0128123	-0.0113930	-0.0211350	6.117267
602	black walnut	7127	-0.0224339	-0.0217011	-0.0336149	4.590613
73	western larch	6745	-0.0181307	-0.0113517	-0.0601380	6.859932
694	swamp tupelo	6463	-0.0277557	-0.0226553	-0.0309769	4.357158
931	sassafras	6422	-0.0250593	-0.0347368	-0.0341496	3.976255
402	bitternut hickory	6368	-0.0222473	-0.0175005	-0.0335616	4.911577
820	laurel oak	6310	-0.0314876	-0.0169942	-0.0264445	5.328190
741	balsam poplar	5921	-0.0411080	-0.0243438	-0.0488560	4.077566
408	black hickory	5835	-0.0273219	-0.0185979	-0.0159591	3.982162

SPCD	CommonName	n	b0	b1	b2	rmse
591	American holly	5828	-0.0254252	-0.0177528	-0.0081532	3.555981
211	redwood	5654	-0.0069422	-0.0137845	-0.0244647	8.609309
462	hackberry	5598	-0.0224973	-0.0203155	-0.0181286	3.949279
491	flowering dogwood	5497	-0.0358367	-0.0422236	-0.0128243	2.628200
351	red alder	5474	-0.0139124	-0.0274425	-0.0218848	6.668121
391	American hornbeam, musclewood	5168	-0.0350985	-0.0253105	-0.0101358	3.512295
653	sweetbay magnolia	5129	-0.0191203	-0.0310973	-0.0315241	4.283565
809	northern pin oak	5089	-0.0317002	-0.0139114	-0.0297109	4.553559
20	California red fir	5033	-0.0140373	-0.0095087	-0.0091706	7.950852

We were able to make base rate estimates for 78 species. The distributions of the estimates for these species are shown below.

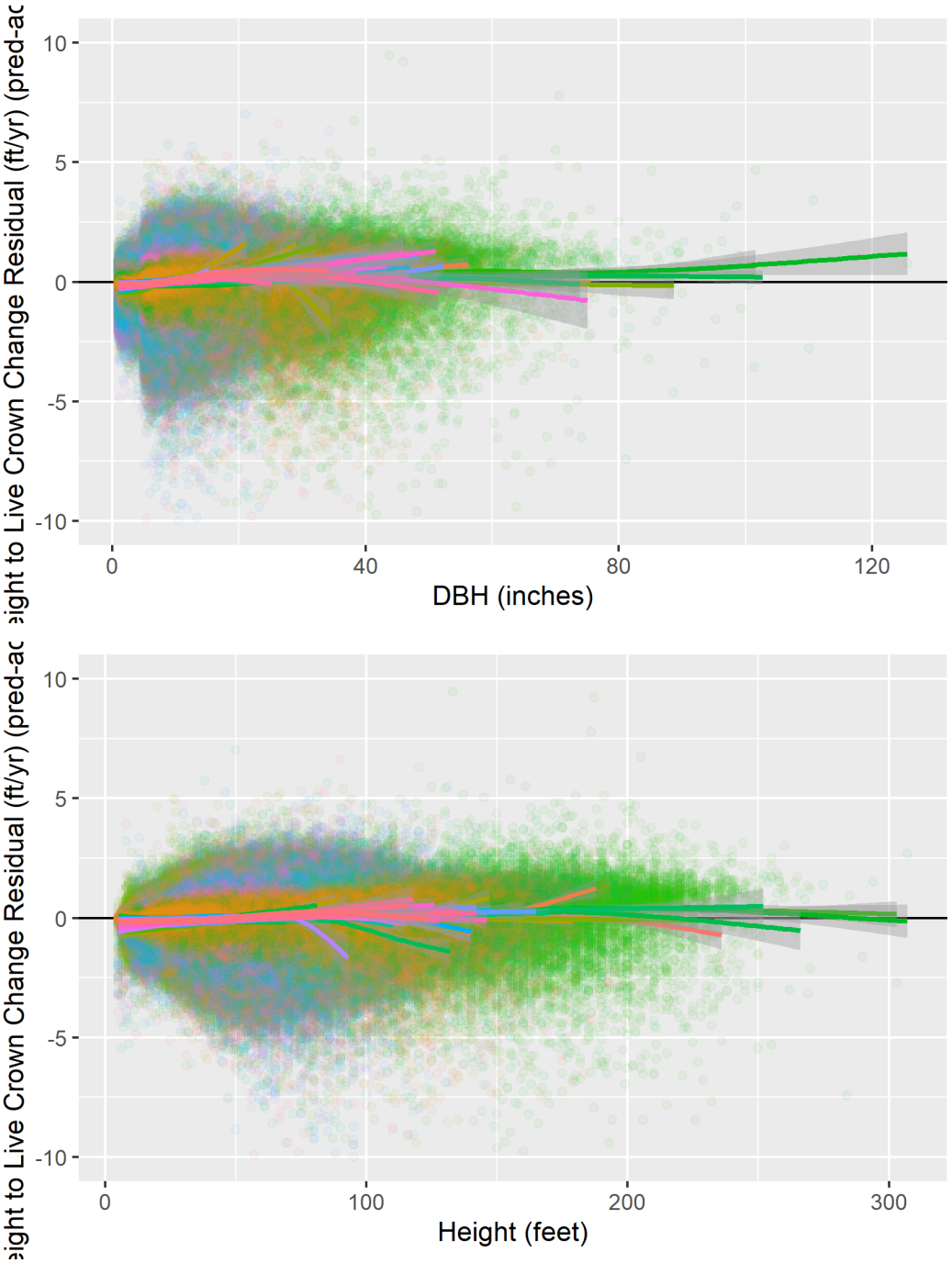


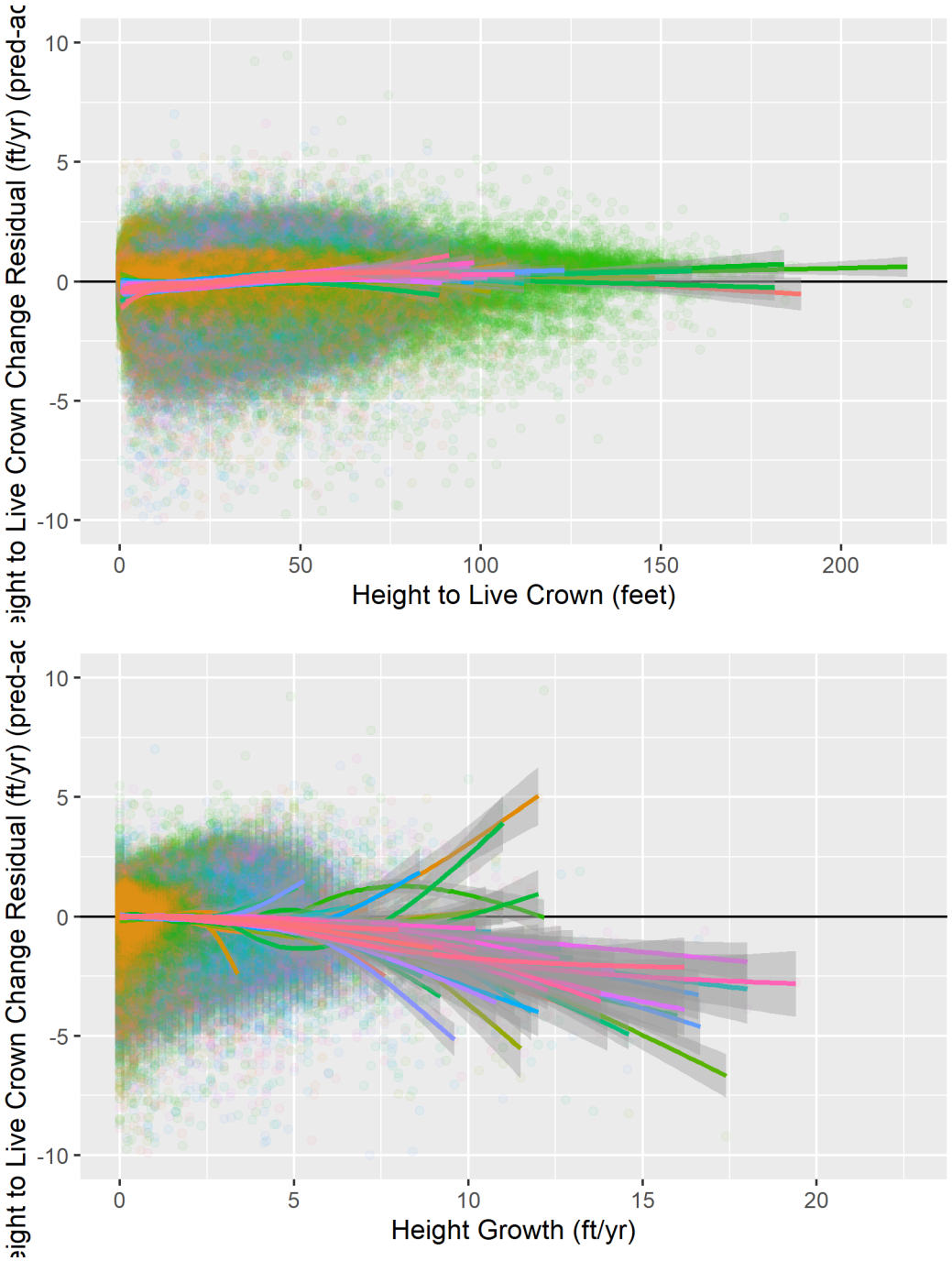

All Species
Conifer
Hardwood
count
8
6
4
2
0
-0.04
-0.03
-0.02
-0.01
0.00
B1

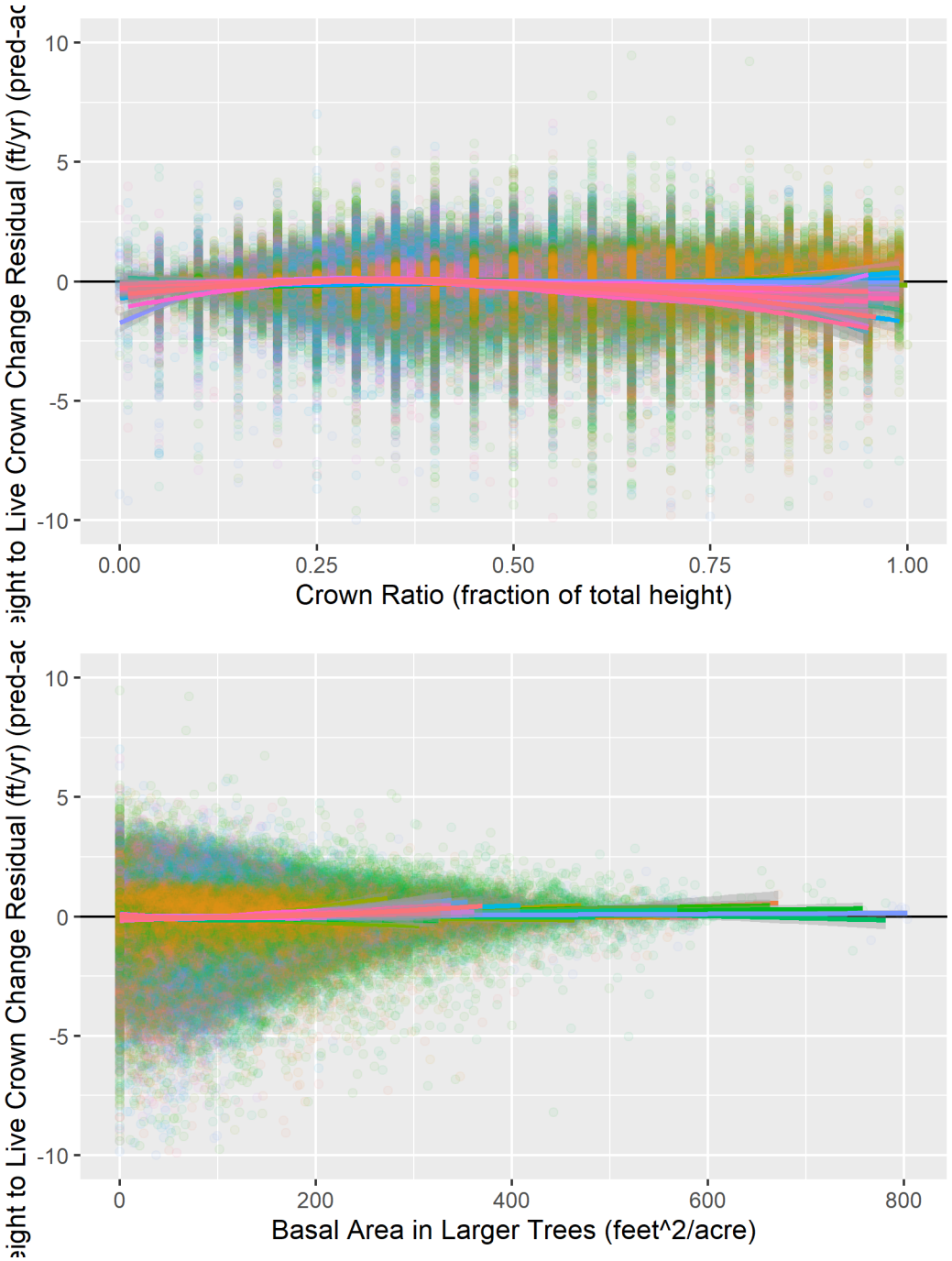

All Species
Conifer
Hardwood
count
6
4
2
0
-0.08
-0.06
-0.04
-0.02
0.00
B2

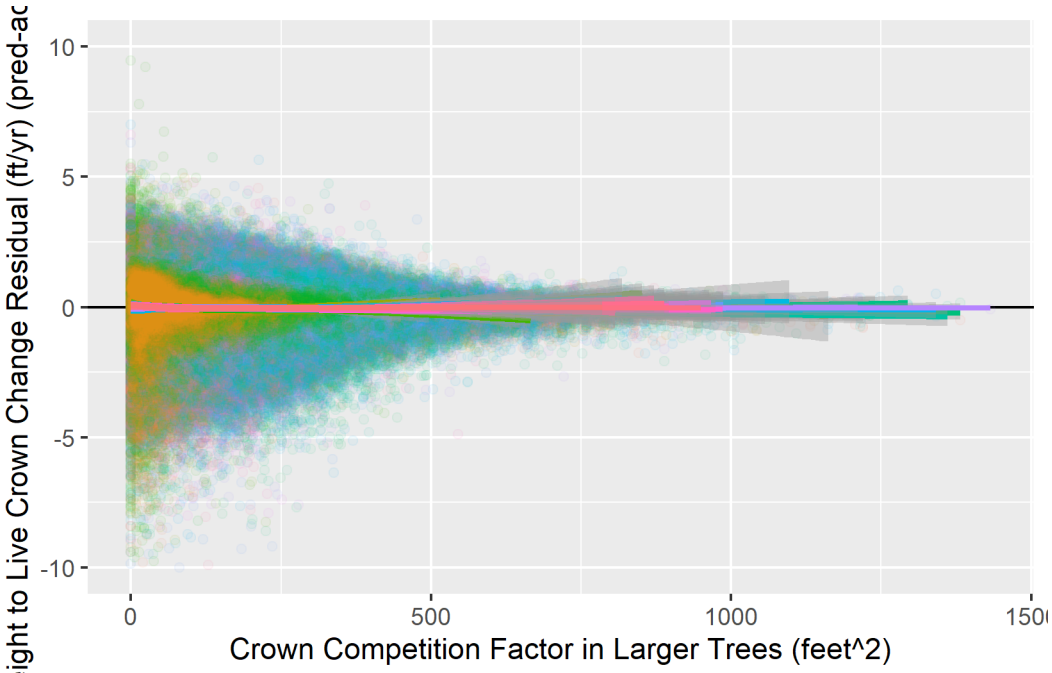
Height to Live Crown Residuals

The following residual graphs show the errors for all fitted species using the proposed model.

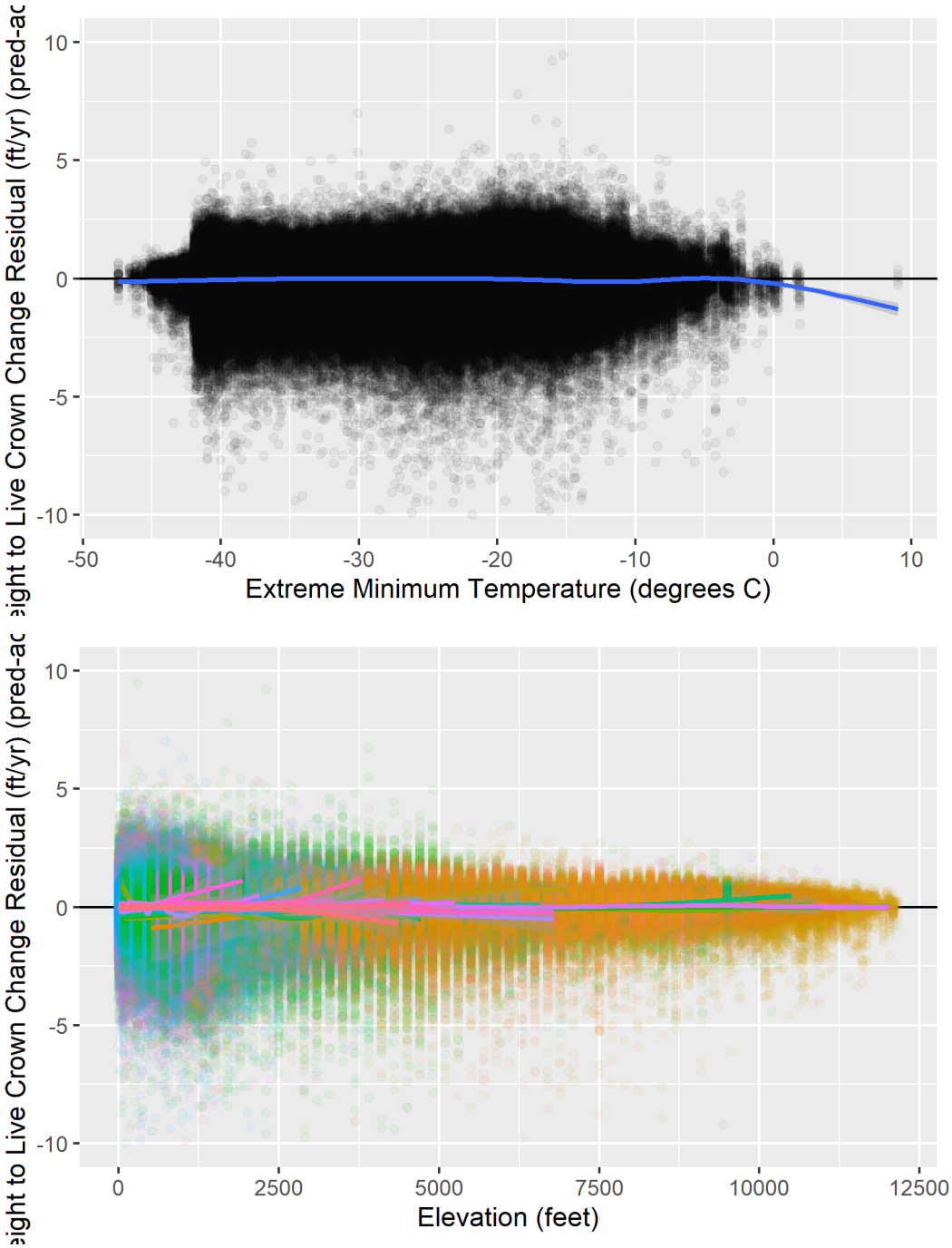


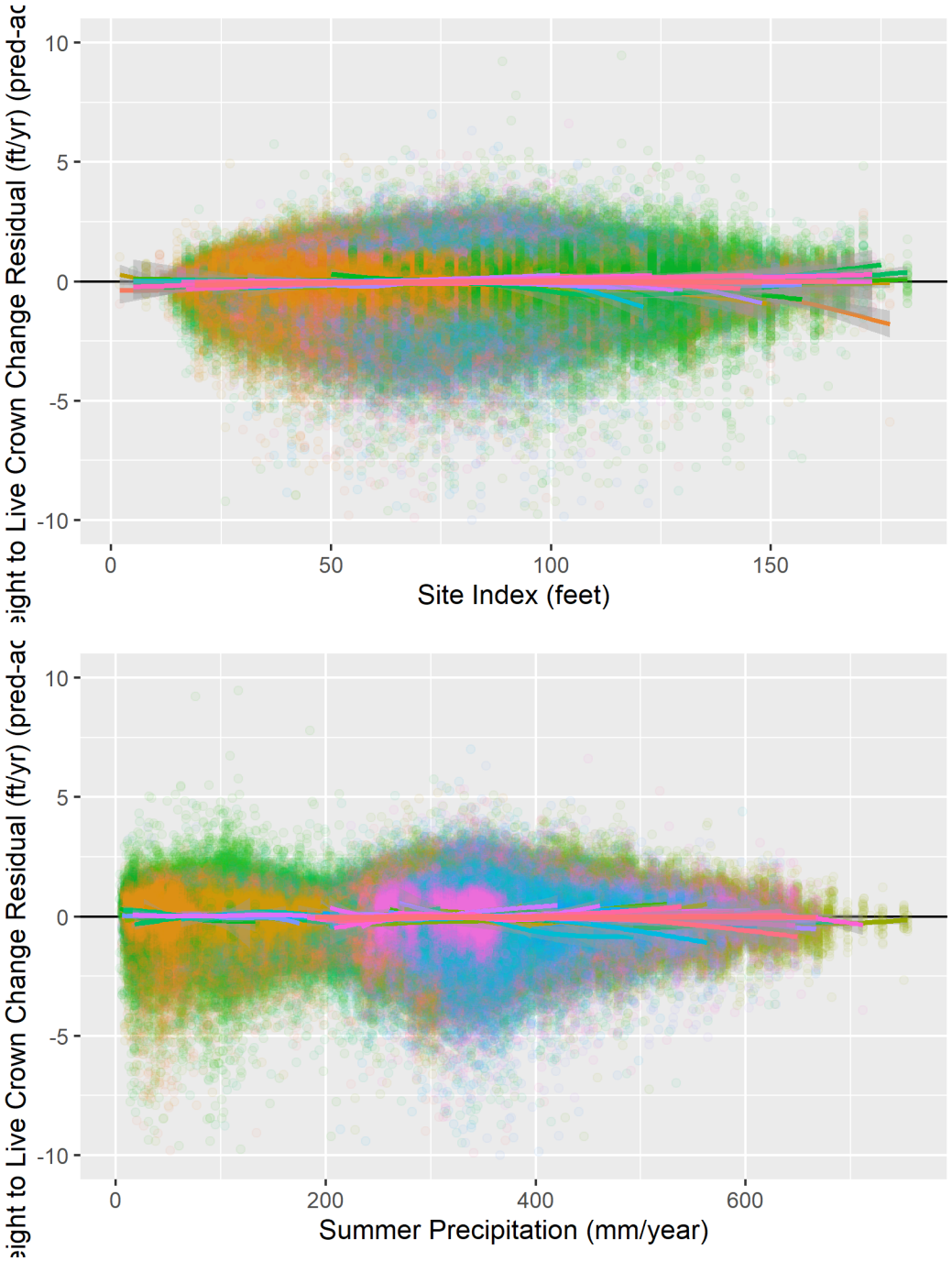

10
5
0
-5
-10
0
50
100
150
200
Height to Live Crown (feet)
10
5
0
-5
-10
0
5
10
15
20
Height Growth (ft/yr)

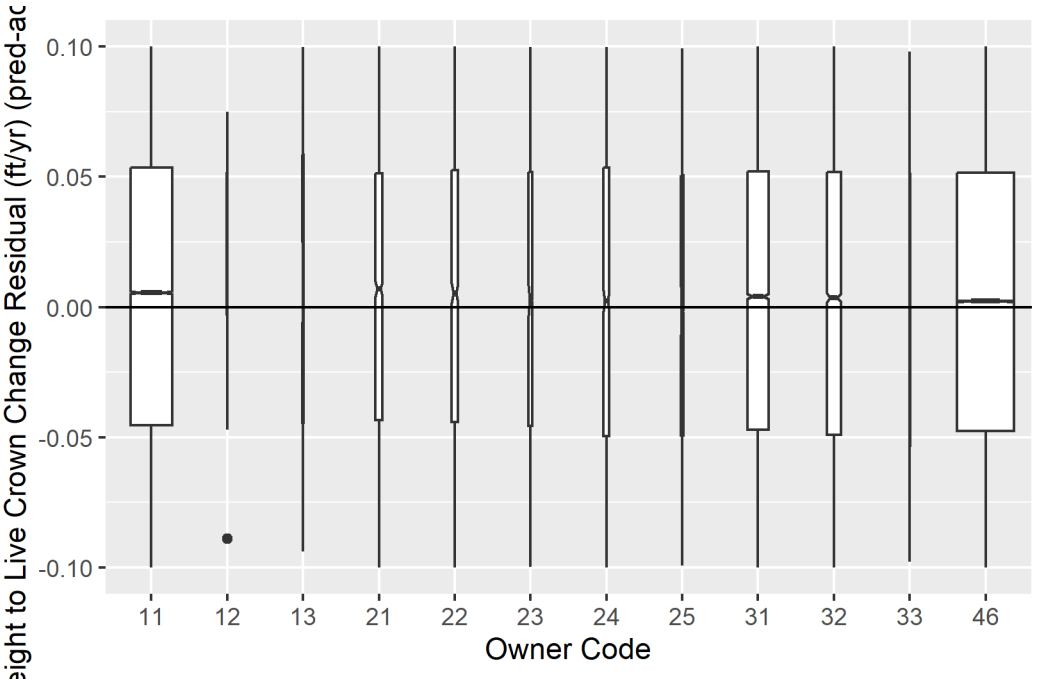

eight to Live Crown Change Residual (ft/yr) (pred-ac
10
5
0
-5
-10
0.00
0.25
0.50
0.75
1.00
Crown Ratio (fraction of total height)
eight to Live Crown Change Residual (ft/yr) (pred-ac
10
5
0
-5
-10
0
200
400
600
800
Basal Area in Larger Trees (feet^2/acre)

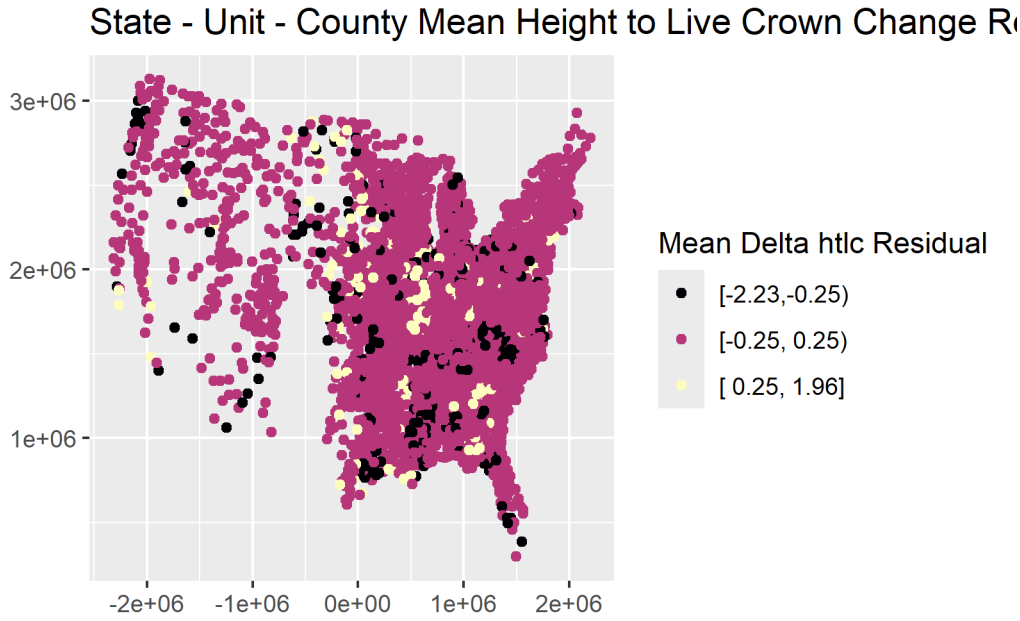

10
5
0
-5
-10
0
500
1000
Crown Competition Factor in Larger Trees (feet^2)


10
5
0
-5
-10
0
2
4
Crown Closure at Tree Tip (fraction of acre)


eight to Live Crown Change Residual (ft/yr) (pred-ac
10
5
0
-5
-10
-50
-40
-30
-20
-10
0
10
Extreme Minimum Temperature (degrees C)
eight to Live Crown Change Residual (ft/yr) (pred-ac
10
5
0
-5
-10
0
2500
5000
7500
10000
12500
Elevation (feet)


eight to Live Crown Change Residual (ft/yr) (pred-ac
10
5
0
-5
-10
0
50
100
150
Site Index (feet)
eight to Live Crown Change Residual (ft/yr) (pred-ac
10
5
0
-5
-10
0
200
400
600
Summer Precipitation (mm/year)


eight to Live Crown Change Residual (ft/yr) (pred-ac
0.10
0.05
0.00
-0.05
-0.10
11
12
13
21
22
23
24
25
31
32
33
46
Owner Code


State - Unit - County Mean Height to Live Crown Change R
3e+06
2e+06
1e+06
-2e+06
-1e+06
0e+00
1e+06
2e+06
Mean Delta htlc Residual
[-2.23,-0.25)
[-0.25, 0.25)
[0.25, 1.96]